

6 Clinical Relevance of Genomics Defined WHO5 Subtypes of Pediatric B-ALL in the Context of Measurable Residual Disease–Directed Risk-Based Therapy

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ABSTRACT

PURPOSE WHO5 (2022) classification of B-lymphoblastic leukemia (B-ALL) incorporates several novel entities requiring high-throughput sequencing for their accurate characterization. The clinical relevance of this classification in the context of contemporary measurable residual disease (MRD)–directed therapy is unclear.

METHODS We analyzed 533 pediatric B-ALL uniformly treated with Indian Collaborative Childhood Leukaemia group (ICiCLE)–ALL-14 protocol as defined by WHO-2016 and reclassified them as per WHO5 using targeted sequencing, FISH, and cytogenetics.

RESULTS Subtype-defining genomic abnormalities were identified in 81.2% of the cohort as per the WHO5 classification. Among the new subtypes, *PAX5*^{alt} and *MEF2D*–r were associated with a trend toward an inferior 3-year event-free survival (EFS) of 32.8% ($P = .003$) and 33.7% ($P = .091$), respectively. We developed a three-tier genomic risk stratification model incorporating 15 genomic subtypes and the *IKZF1* deletion. Children with standard (SGR), intermediate (IGR), and high genomic risk (HGR) demonstrated 3-year EFS of 80.4%, 59.3%, and 45.8% ($P < .0001$), and 3-year overall survival of 89.6%, 75.3%, and 62.3% ($P < .0001$), respectively. Genomic risk further identified heterogeneous outcomes among ICiCLE risk groups ($P < .0001$). SGR was associated with superior EFS irrespective of MRD status (3-year EFS 80.5% in postinduction [PI] MRD–negative v 80.8% PI-MRD–positive patients, $P = .530$). On multivariable analysis, genomic risk (hazard ratio [HR], 1.7 [95% CI, 1.41 to 2.01]; $P < .0001$), initial ICiCLE risk (HR, 1.3 [95% CI, 1.06 to 1.49]; $P = .009$), and PI-MRD (HR, 2.2 [95% CI, 1.66 to 2.90]; $P < .0001$) independently predicted EFS.

CONCLUSION The study demonstrates the potential role of genomic risk stratification, in conjunction with MRD, in stratifying patients into clinically relevant risk categories.

ACCOMPANYING CONTENT

 [Data Sharing Statement](#)

 [Data Supplement](#)

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INTRODUCTION

In the past few decades, successful adoption of risk-directed therapy has resulted in remarkably improved pediatric B-lymphoblastic leukemia/lymphoma (B-ALL) outcomes. Assessment of therapeutic response by detecting and quantitating measurable residual disease (MRD) forms the cornerstone of contemporary risk-adapted protocols.

The underlying biology of the B-ALL subtype, which includes abnormalities of chromosome number such as high hyperdiploid (HeH) as well as chimeric gene fusions such as *ETV6*:

RUNX1, are clinically relevant even in the context of MRD-directed therapies.¹ Contemporary protocols have thus adopted an integrated approach incorporating both baseline genomics and MRD as an optimal strategy for risk-directed therapy.^{1,2}

In the past decade, advances in genome sequencing technologies such as whole-genome and whole-transcriptome sequencing have substantially improved our understanding of B-ALL biology, leading to the identification of several novel biologic subgroups.^{3,4} The WHO5 and ICC classifications have incorporated several of these new subtypes, such as *TCF3::HLF1*, *PAX5* alteration (*PAX5*^{alt}), *PAX5* P80R,

CONTEXT

Key Objective

To evaluate the clinical relevance of the WHO5 (2022) genomic classification of pediatric B-lymphoblastic leukemia (B-ALL) in the context of measurable residual disease (MRD)-directed therapy and to determine whether genomic subtypes can provide additional prognostic information beyond established clinical risk stratification.

Knowledge Generated

A three-tier genomic risk model (standard, intermediate, and high genomic risk) demonstrated significant differences in event-free survival and overall survival. Genomic risk also improved prognostic discrimination beyond Indian Collaborative Childhood Leukaemia group clinical risk and postinduction MRD.

Relevance

These findings show that WHO5 genomic classification is feasible in routine practice using targeted sequencing and provides clinically relevant prognostic information for risk-adapted therapy in pediatric B-ALL.

DUX4 rearrangement (*DUX4*-r), *MEF2D*-r, *ZNF384*-r, *NUTM1*-r, *ETV6::RUNX1*-like, *IGH::MYC*.^{5,6} Recently, Jha et al have shown the clinical significance of the new subtypes of B-ALL in the context of the MRD-directed St Jude Total Therapy Study 16 protocol.⁷ It is important to validate and incorporate these genomics data into clinical practice for prognostication and targeted therapy.^{7,8}

A major challenge in detecting the WHO5-recognized subtypes of B-ALL in routine practice is the heterogeneous biology of B-ALL, which requires integration of copy-number alterations, fusions involving multiple genes with various partners, and somatic mutations to comprehensively detect all subtypes. For diagnosis of these new entities, legacy workflows such as fluorescent in situ hybridization (FISH), multiplex polymerase chain reaction (PCR), and multiplex ligation-dependent probe amplification require multiple rounds of reflexive testing that would result in time delay. Furthermore, PCR-based tests require knowledge of both partners involved in the fusion, which is not always feasible, especially with the novel described entities.

Whole-transcriptome sequencing (WTS) has been proposed as a one-stop solution for this problem; however, its clinical implementation remains challenging because of cost, infrastructure requirements, and turnaround time, particularly when results are needed to guide therapeutic decisions. We have previously validated NARASIMHA, a laboratory-developed, cost-effective targeted RNA-sequencing assay to detect chimeric gene fusions with knowledge of only one partner gene (eg, *ABL1*). We therefore hypothesized that a stepwise approach integrating legacy techniques with targeted DNA and RNA sequencing would enable identification of the majority of WHO5-defined B-ALL entities.

Here, we present the clinical utility of this approach in detecting WHO5 subtypes in a large, uniformly treated cohort of pediatric B-ALL. We further describe the clinical

relevance of such an approach in predicting outcomes in the context of MRD-directed therapy in childhood B-ALL.

METHODS

Diagnosis

A total of 533 newly diagnosed pediatric B-ALL, age 1–15 years, enrolled in the Indian Collaborative Childhood Leukaemia group (ICiCLE)-ALL-14 trial (Clinical Trials Registry-India number, CTRI/2015/12/006,434), from whom peripheral blood or bone marrow samples were received for molecular testing between January 2019 to March 2022, were included in this study (Fig 1). B-ALL diagnosis was made using 10–12 color, 5-tube multicolor flow cytometry (MFC), as described elsewhere.⁹ Also, ploidy status was determined using the FxCycle Violet dye-based MFC method.¹⁰ Institutional Ethics Committee of ACTREC, Tata Memorial Centre provided ethical approval for this work (IEC III—900924). Informed consent was taken from the legal guardians while enrolling in the ICiCLE-ALL-14 trial (Clinical Trials Registry-India number, CTRI/2015/12/006,434) for participation and data sharing.

Cytogenetics Workup

FISH was performed for *BCR::ABL1*, *TCF3::PBX1*, *ETV6::RUNX1*, *KMT2A*-r, *IKZF1* deletions, and centromeric probes for chromosomes 4, 10, and 17. Ploidy analysis was also performed on Giemsa-stained metaphases as previously described.¹¹ Detailed probe sets and methodology are provided in the Data Supplement.

Polymerase Chain Reaction–Based Screening

Common translocations (*BCR::ABL1* [E13/14A2 and E1A2], *ETV6::RUNX1*, *TCF3::PBX1*, and *KMT2A::AFF1*) were screened using real-time PCR. *IKZF1* deletions were detected using fragment length analysis.^{12,13}

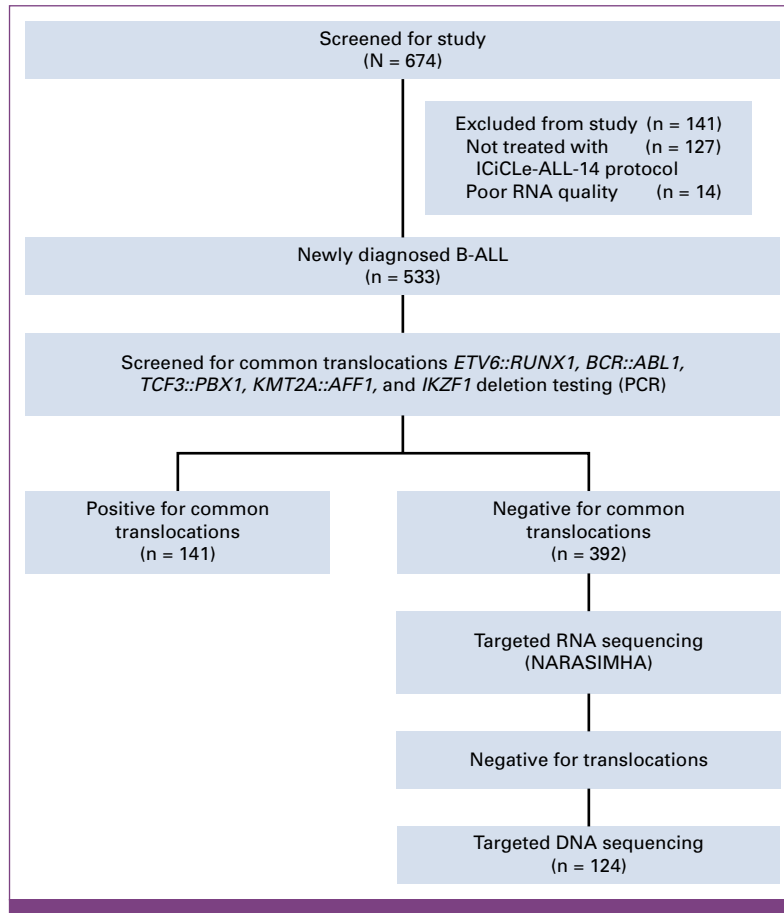


FIG 1. Flow diagram indicating testing strategy implemented for B-ALL patient evaluation. ICiCle, Indian Collaborative Childhood Leukaemia group; PCR, polymerase chain reaction.

Next-Generation Sequencing

Targeted RNA sequencing was performed in 392 patients (73.5%) who were negative for common translocations (Fig 1, Data Supplement, Table S1). Patients with negative RNA sequencing (n = 124) were subjected to targeted DNA sequencing. The details of these assays and bioinformatics analysis can be seen in the Data Supplement.

Risk Stratification and Treatment Protocol

Patients were stratified and treated according to the ICiCle-ALL-14 risk stratification detailed in the Data Supplement (Fig S1).¹⁴ ICiCle's final risk stratification differs from initial risk stratification primarily by incorporating postinduction (PI) MRD into the assessment. PI and postconsolidation (PC) MRD assessments were performed using our previously validated highly sensitive MFC assay.¹⁵

Classification According to WHO-2016 and WHO5 (2022)

The pediatric B-ALL cases were assigned to defined genomic categories as per the WHO-2016 Classification and WHO5

Classification of Tumours of Haematopoietic and Lymphoid Tissues integrating DNA ploidy by MFC, FISH, chromosomal counting, screening PCR, NARASIMHA, and targeted DNA sequencing.⁵

Clinical End Points and Statistical Analysis

Overall survival (OS) was defined from diagnosis to last follow-up or death. Event-free survival (EFS) was defined from diagnosis to death, relapse, or refractory status. Cumulative incidence of relapse (CIR) was calculated from diagnosis to last follow-up or relapse, with death due to any other cause treated as competing event. Associations between categorical variables were assessed using the chi-square or Fisher's exact test. Survival outcomes were estimated using the Kaplan-Meier method and compared using the log-rank test. Group comparisons were performed using the log-rank test, and hazard ratio (HR) with 95% CI were estimated using the Cox proportional hazards model. The detailed statistical analysis for genomic sub-groups, genomic stratification, multivariable analysis (MVA), and prognostic modeling is given in the Data Supplement.

RESULTS

Clinical, Laboratory, and Risk Stratification Characteristics of Patients Enrolled

Our final analysis included 533 children with B-ALL (male:female ratio 1.99:1; median age 5 years). According to the National Cancer Institute risk stratification, most patients were classified as standard risk (SR, 67.2%). However, with the final ICiCLE risk stratification, the majority (57.4%) were categorized as high risk (HiR), followed by SR (22.9%) and intermediate risk (IR, 19.7%). PI-MRD positivity ($\geq 0.01\%$) was observed in 32.2% of patients (167/519), and 9.8% (24/244) remained positive PC. A summary of clinical, laboratory, and genomic characteristics, as well as MRD responses, can be seen in the Data Supplement (Table S2).

Characterization of Genomic Subgroups and Their Correlations

Through the integration of conventional PCR, targeted RNA sequencing, cytogenetics, and immunophenotyping, we identified 13 distinct genomic subgroups within our cohort (Data Supplement, Table S3). *PAX5*^{alt} (24, 4.5%), *BCR::ABL1*-like (23, 4.3%), *DUX4-r* (12, 2.3%), *MEF2D-r* (13, 2.4%), *ZNF384-r* (3, 0.6%), and *TCF3::HLF* (4, 0.7%) subgroups of B-ALL were exclusively identified using RNA sequencing (Fig 2). The details of fusion partners in each genomic subgroup are given in the Data Supplement. Targeted DNA sequencing further helped identify additional *PAX5*^{alt} (n = 4) and the *PAX5* P80R group (n = 4). Using this approach, we could identify a genomic abnormality in 81.2% of all cases when compared with legacy approaches (61.7%). The MRD characteristics of each genomic subgroup are detailed in the Data Supplement (Fig S2). *IKZF1* deletion was identified in 69 (15.1%) of the 458 patients tested and showed a high prevalence in the *BCR::ABL1* (56.5%) and *BCR::ABL1*-like (55.6%) genomic subgroups (Data Supplement, Fig S3).

Restratification From the WHO-2016 to WHO5 Classification

Classification remained unchanged for 349 patients (*ETV6::RUNX1*, HeH, *BCR::ABL1*, *TCF3::PBX1*, *KMT2A-r*, hypodiploidy, and *iAMP21*) when reclassified according to WHO5 compared with WHO-2016. Among WHO-2016-defined not otherwise specified (NOS) subgroup (n = 184, 34.5% of the entire cohort), 84 (45.6%) children were reclassified into newly recognized subtypes (Fig 2).

Patient Outcomes

The 3-year EFS and OS of the cohort were 64.9% and 78.7%, respectively. At a median follow-up of 38.9 months, the median EFS was 63.7 months (95% CI, 56.7 to 66.5), while median OS was not reached. PI-MRD positivity predicted a

significantly inferior EFS (HR, 2.0 [95% CI, 1.50 to 2.77]; $P < .0001$) and OS (HR, 2.1 [95% CI, 1.42 to 3.20]; $P < .0001$). Similarly, PC-MRD positivity was associated with markedly worse EFS (HR, 19.0 [95% CI, 3.92 to 91.85]; $P < .0001$) and OS (HR, 2.4 [95% CI, 0.89 to 6.63]; $P = .01$; Data Supplement, Fig S4). The impact of ICiCLE risk stratification on patient outcomes is detailed in the Data Supplement (Fig 3 and Data Supplement, Fig S5).

Outcomes for Individual Genomic Subgroups

Evaluation of genomic subgroups, acknowledging the limited number of cases in several categories, showed trend toward inferior EFS in *PAX5*^{alt} (HR, 2.2 [95% CI, 1.32 to 3.81]; log-rank $P = .003$), *BCR::ABL1* (HR, 1.8 [95% CI, 1.00 to 2.75]; log-rank $P = .050$), and *MEF2D-r* (HR, 1.8 [95% CI, 0.91 to 3.55]; log-rank $P = .091$; Fig 4, Data Supplement, Table S4). Similarly, with respect to OS, a trend toward inferior OS was observed in *PAX5*^{alt} (HR, 3.4 [95% CI, 1.85 to 6.32]; log-rank $P < .0001$; Fig 5). Subtype-independent analysis demonstrated inferior EFS in cases with *IKZF1* deletion (HR, 1.7 [95% CI, 1.18 to 2.33]; log-rank $P = .004$). Among the four children with *TCF3::HLF*-positive B-ALL, three had refractory disease and one experienced early relapse, suggesting a trend toward poorer outcomes; however, the small sample size precluded formal statistical analysis and definitive interpretation.

Conversely, HeH and *ETV6::RUNX1* were associated with superior EFS (HR, 0.7 [95% CI, 0.53 to 0.94]; $P = .018$ and HR, 0.7 [95% CI, 0.44 to 0.98]; $P = .042$, respectively; Data Supplement, Table S4). Kaplan-Meier curves depicting EFS and CIR for each genomic subgroup are shown in Figure 5.

Genomic Risk Assignment

We devised a three-tier genomic risk model (standard [SGR], intermediate [IGR], and high [HGR]) incorporating variables found to be significantly predictive of EFS on univariate analysis in this study. We derived two HR cutoffs to define three genomic risk tiers using the distribution of Cox model effect sizes as detailed in the Data Supplement. The details of this classification can be seen in Figure 6. The 3-year EFS for SGR, IGR, and HGR was 80.4%, 59.3%, and 45.8%, respectively ($P < .0001$), and the 3-year OS for SGR, IGR, and HGR was 89.6%, 75.3%, and 62.3%, respectively ($P < .0001$; Data Supplement, Table S5). The 3-year CIR for SGR, IGR, and HGR was 20.0%, 38.2%, and 48.8%, respectively ($P < .0001$).

Genomic Risk Identifies Heterogeneous Outcomes Within ICiCLE Risk Groups

As the treatment intensity of the ICiCLE initial SR and IR is similar, these two groups were combined for this analysis, while the HiR group was analyzed separately. Genomic risk stratification retained its prognostic significance

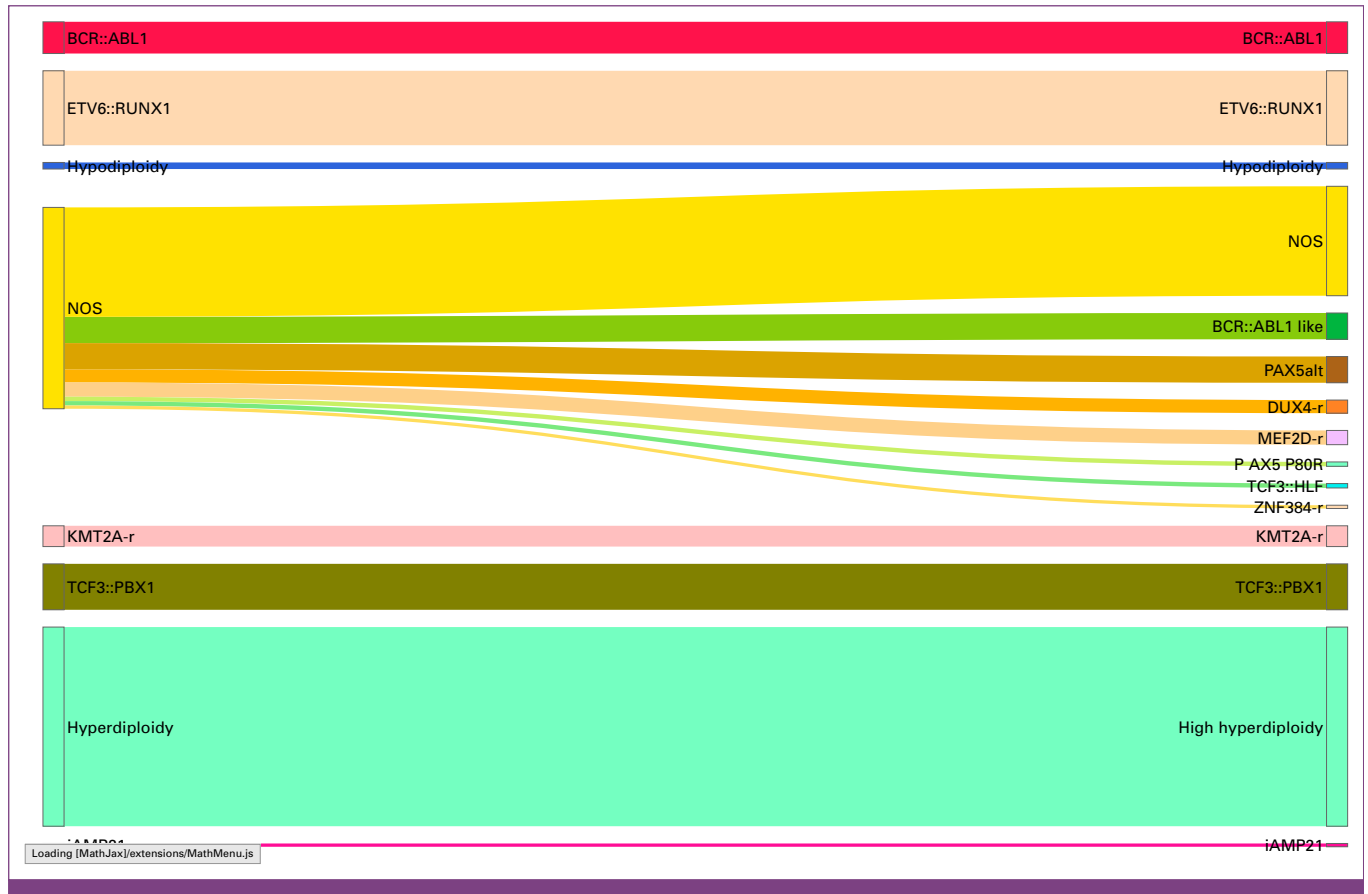


FIG 2. Sankey diagram illustrating the reclassification of B-ALL cases from WHO 2016 to WHO5 (2022) B-ALL classification. NOS, not otherwise specified.

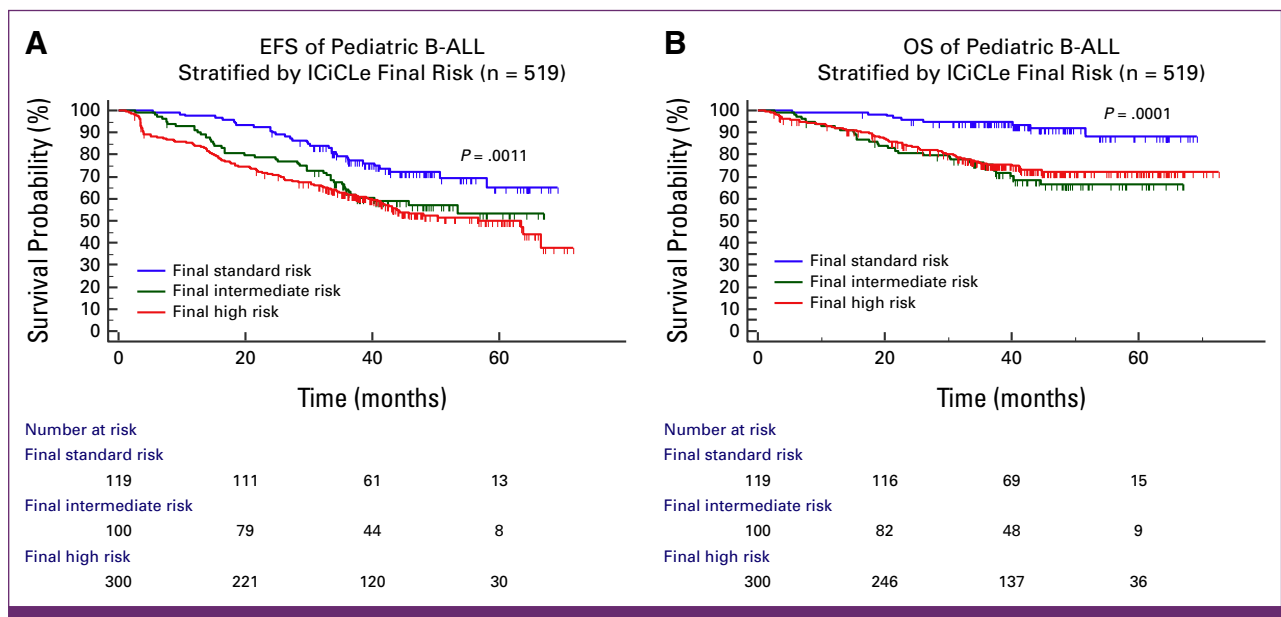


FIG 3. (A) EFS stratified by ICIcLe final risk; (B) OS stratified by ICIcLe final risk. EFS, event-free survival; ICIcLe, Indian Collaborative Childhood Leukaemia group; OS, overall survival.

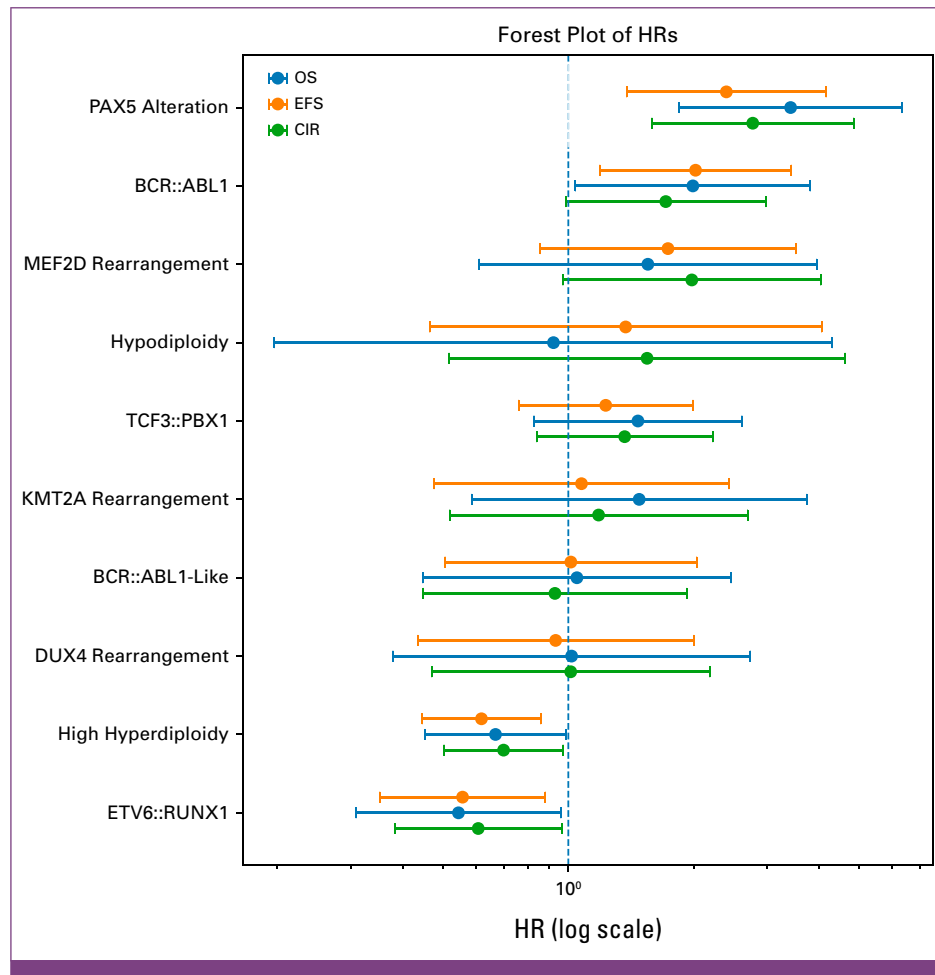


FIG 4. Forest plot illustrating HRs with 95% CIs for EFS, OS, and CIR across various genomic subgroups of pediatric B-cell ALL. CIR, cumulative incidence of relapse; EFS, event-free survival; HR, hazard ratio; OS, overall survival.

within both ICiCLE initial SR/IR and HiR groups. The 3-year EFS of SGR, IGR, and HGR within ICiCLE initial SR/IR was 81.0%, 62.5%, and 48.5%, respectively ($P < .0001$), and that within ICiCLE initial HiR was 78.6%, 51.3%, and 43.3%, respectively ($P < .0001$; **Fig 7A**). Similarly, the 3-year OS of SGR, IGR, and HGR within ICiCLE initial SR/IR was 90.7%, 77.3%, and 62.6%, respectively ($P < .0001$), and that within ICiCLE initial HiR was 86.8%, 69.7%, and 62.1%, respectively ($P = .005$; Data Supplement, Table S5).

Genomic Risk Stratification Reveals MRD-Independent Favorable Outcomes in the SGR Group

Patients in the SGR group exhibited superior outcomes regardless of PI-MRD status, with 3-year EFS rates of 80.5% for MRD-positive and 80.8% for MRD-negative patients ($P = .530$; **Figure 7B**), and 3-year OS rates of 91.0% and 87.2%, respectively ($P = .253$). By contrast, MRD status significantly influenced outcomes within IGR and HGR groups as detailed in the Data Supplement.

Multivariable Analysis

On MVA, genomic risk stratification (HR, 1.7 [95% CI, 1.41 to 2.01]; $P < .0001$), initial ICiCLE risk (HR, 1.3 [95% CI, 1.06 to 1.49]; $P = .009$), and PI-MRD (HR, 2.2 [95% CI, 1.66 to 2.90]; $P < .0001$) independently predicted EFS. Similarly, for OS, genomic risk (HR, 1.7 [95% CI, 1.31 to 2.09]; $P = .0001$), PI-MRD positivity (HR, 2.1 [95% CI, 1.44 to 3.04]; $P < .0001$), and initial ICiCLE risk (HR, 1.3 [95% CI, 1.02 to 1.61]; $P = .032$) remained independent predictors (Data Supplement, Table S6). The details of MVA for individual genomic subtypes are given in the Data Supplement.

Genomic Risk Provides Incremental Prognostic Information

In nested Cox models for EFS, a baseline model incorporating ICiCLE initial risk and PI-MRD showed modest discrimination (C-index 0.63; 95% CI, 0.60 to 0.68). Addition of genomic risk based on the WHO-2016 classification

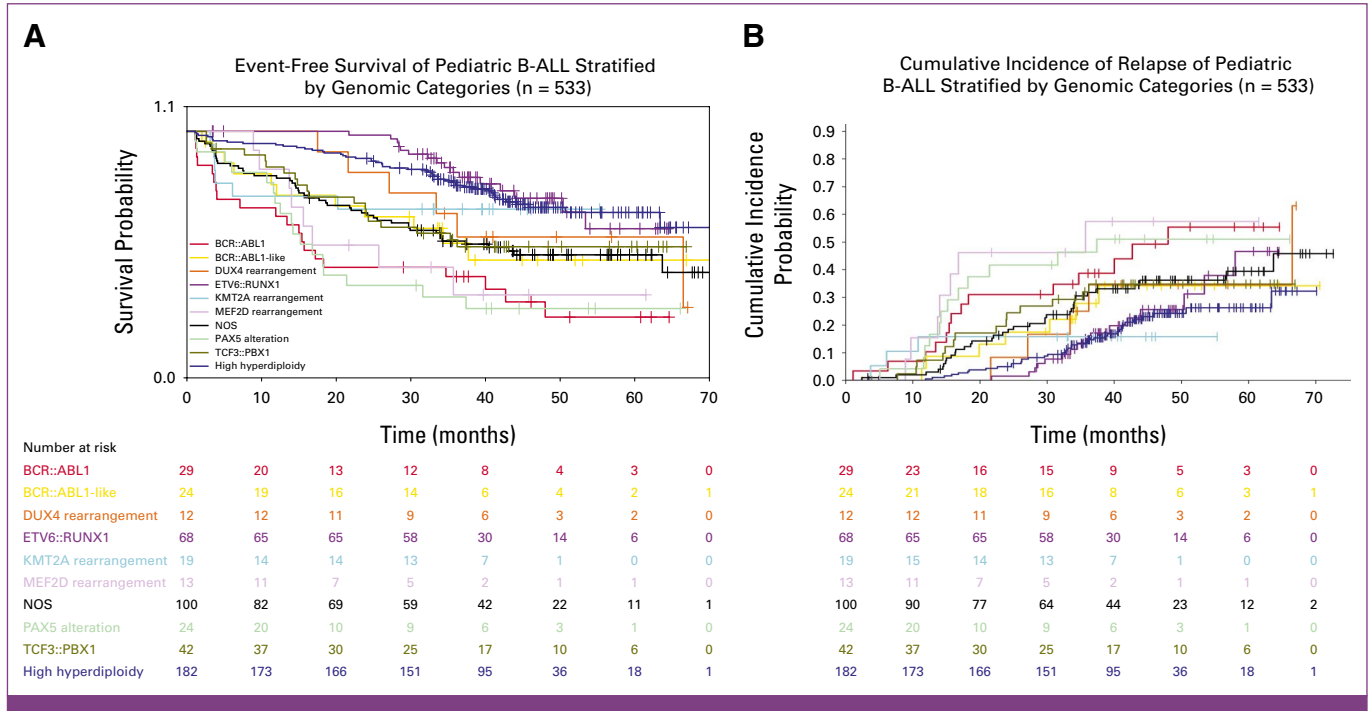


FIG 5. (A) EFS of patients with pediatric B-ALL (n = 533) categorized by genomic subtypes; (B) CIR of patients with pediatric B-ALL (n = 533) categorized by genomic subtypes. CIR, cumulative incidence of relapse; EFS, event-free survival; NOS, not otherwise specified.

improved discrimination (C-index 0.66; 95% CI, 0.62 to 0.70), and the WHO5 classification further improved discrimination (C-index 0.69; 95% CI, 0.65 to 0.73), indicating incremental prognostic value.

DISCUSSION

Pioneering recent studies have used unbiased broad-spectrum approaches such as WTS/whole-genome sequencing (WGS)

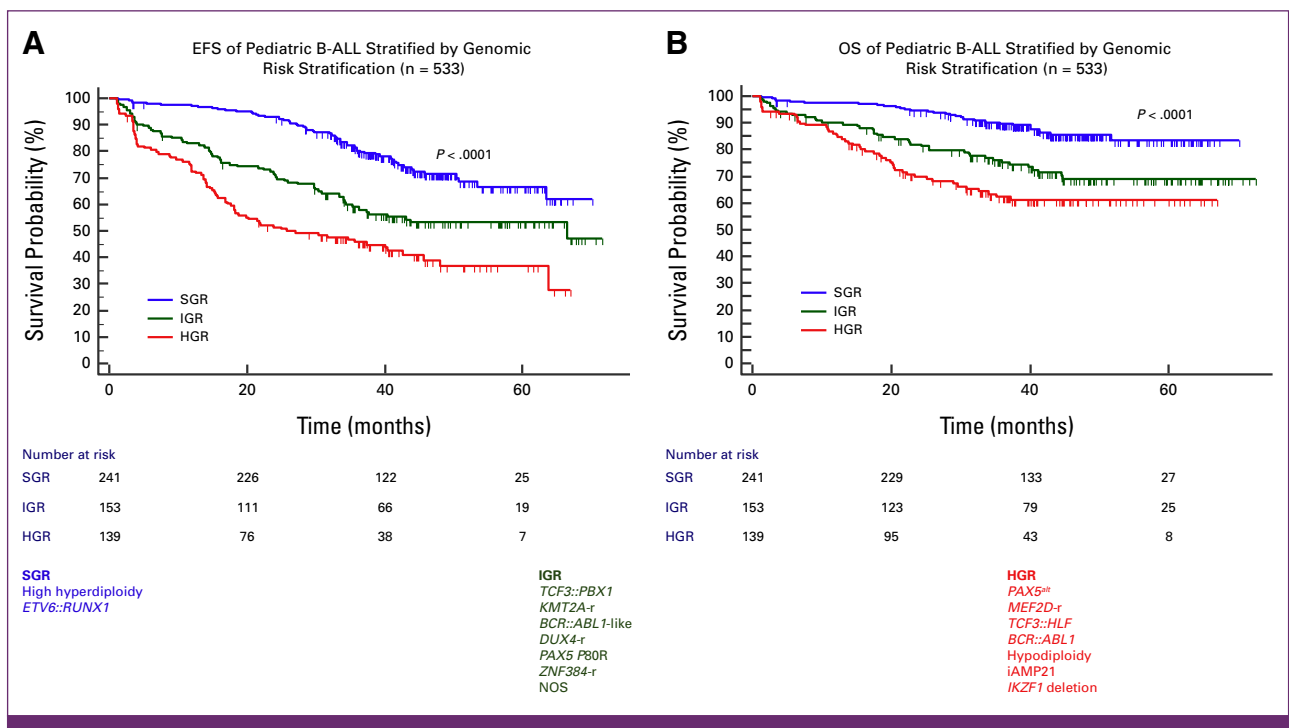


FIG 6. (A) EFS and (B) OS of patients with pediatric B-ALL stratified by genomic risk classification into SGR, IGR, and HGR groups. EFS, event-free survival; IGR, intermediate genomic risk; HGR, high genomic risk; OS, overall survival; SGR, standard genomic risk.

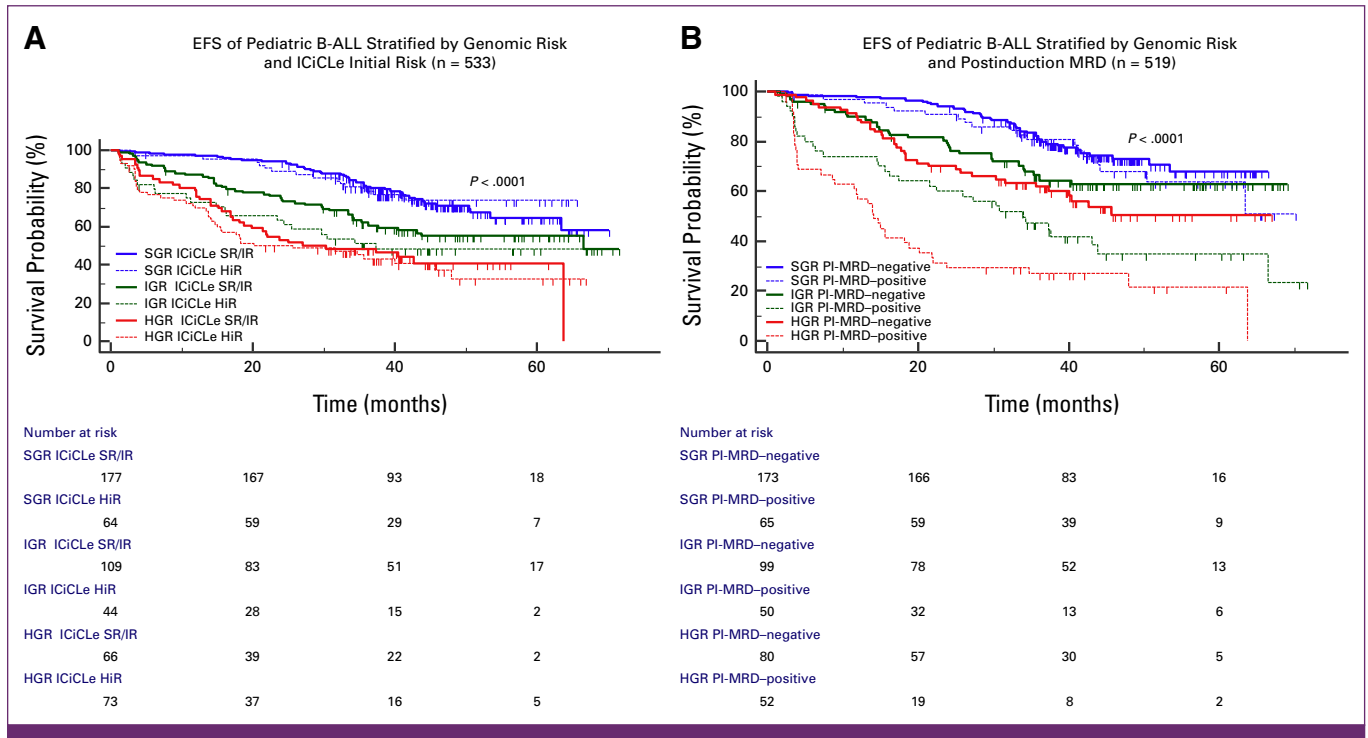


FIG 7. (A) EFS stratified by ICiCLE initial risk stratification and genomic risk reveals overlap between intermediate and high ICiCLE risk groups. (B) EFS stratified by genomic risk and postinduction MRD. EFS, event-free survival; HGR, high genomic risk; HiR, high risk; ICiCLE, Indian Collaborative Childhood Leukaemia group; IGR, intermediate genomic risk; IR, intermediate risk; MRD, measurable residual disease; PI, postinduction; SGR, standard genomic risk; SR, standard risk.

to uncover novel genetic subtypes,^{4,16} which have been recognized as distinct entities in the latest WHO5/ICC classification of B-ALL.⁵ Routine clinical application of WTS/WGS assays remains challenging. Herein, we demonstrate that targeted RNA and DNA sequencing, in combination with standard FISH/cytogenetics, is routinely implementable in clinical practice and can detect majority of the WHO5-recognized subtypes of B-ALL. We demonstrate that a targeted sequencing approach identifies prognostically relevant subgroups even in the era of contemporary MRD-directed therapies.

Among the recently defined genetic categories, we found frequencies of 4.5% for *BCR::ABL1*-like, 4.5% for *PAX5*^{alt}, 2.44% for *MEF2D*-r, 2.25% for *DUX4*-r, 0.75% each for *PAX5* P80R and *TCF3::HLF*, and 0.56% for *ZNF384*-r. The proportions of *PAX5*^{alt} (4.5% v 7.5%)⁴ and *DUX4*-r (2.25% v 4%)³ cases in our cohort are lower than those reported in the discovery cohorts. The NARASIMHA assay used in this study focuses only on detecting chimeric gene fusions. Alternative approaches based on gene expression levels may lead to the discovery of novel phenocopies. Using our approach, we are limited to identifying *ETV6::RUNX1*-like ALL only with known fusion partners, including *BCL2L14*, *BORCS5*, *CREBBP*, *MSH6*, *NID1*, and *PMEL*. A subset of this entity based on gene expression may be missed. However, this is only a rare provisional entity in WHO5, and its prognostic impact remains uncertain.

Another factor to consider is that we report results only on a large cohort of children uniformly recruited in the ICiCLE-ALL-14 trial. Interestingly, transcriptomic sequencing findings from the St Jude Total Therapy Study 16 trial reported similar *BCR::ABL1*-like (3%) and *PAX5*^{alt} (4.85%) cases.⁷ In total, 81.2% of the children analyzed could be assigned to a defined genetic subgroup using our approach and 18.7% of cases remained NOS, compared with 12.1% B-other cases reported in the St Jude Total Therapy Study 16.⁷ Although this gap highlights the need for broader-spectrum genomic profiling in future studies, our approach provides a viable framework that detects most of the WHO5-recognized genetic abnormalities in children with B-ALL in a trial setting.

Here, we developed three-tier genetic risk model incorporating variables predictive of EFS in a large cohort of children treated uniformly with MRD-directed risk-adapted ICiCLE-ALL-14 protocol. For rare genomic subgroups with limited representation in our cohort, risk assignment was informed by outcomes reported in the existing literature. The *HeH* and *ETV6::RUNX1* were included in SGR group. *DUX4*-r has been reported to be associated with superior outcome in earlier reports.^{7,17} In our cohort, children with *DUX4*-r had 3-year EFS of 66.7% (HR, 1.0 [95% CI, 0.48 to 1.97]). Although these cases were categorized under the IGR group based on our HR thresholds, this classification warrants cautious interpretation, given the limited sample size (n = 12), which may underpower statistical significance and

affect reliability of outcome estimates in this subgroup. *KMT2A-r* are often linked with adverse outcomes in both infants and adults with B-ALL. Our study included a cohort of 1–15-year-olds, and similar to the study where noninfant children with *KMT2A-r* were evaluated,¹⁸ our cohort of *KMT2A-r* B-ALL demonstrated an intermediate 3-year EFS of 68.4% (HR, 1.1 [95% CI, 0.50 to 2.21]), thus being grouped into IGR. The *PAX5^{alt}* group had the median EFS of 16.8 months (95% CI of median, 12.5 to 37.4; HR, 2.2 [95% CI, 1.32 to 3.81], log-rank *P* = .003) in our cohort and is included in the HGR category. Initial studies from the St Jude Children's Research Hospital (SJCRH)/Children's Oncology Group (COG) group have reported an intermediate prognosis of *PAX5^{alt}* with MRD-directed therapy^{4,7}; however, based on the trend toward poorer EFS in our cohort, we have included the *PAX5^{alt}* in the HGR category. This finding aligns with the recently published study by Chang et al, where approximately 50% of children with *PAX5^{alt}* ALL relapsed, highlighting the poor prognosis of this subtype in standard-risk pediatric patients.¹⁹ We included *IKZF1* deletion, *MEF2D-r*, and *TCF3::HLF* in the HGR category, which is in line with previous findings by the SJCRH/COG group.^{4,7} We also observed a trend toward inferior outcomes in the *BCR::ABL1*-like ABL-class subgroup. However, because of the small sample size and variability in tyrosine kinase inhibitor (TKI) selection, we did not assign this subgroup separately and retained it within the broader *BCR::ABL1*-like category in IGR. Larger cohorts treated with uniform TKI strategies will be needed to determine its true prognostic significance. Previous reports have shown that although *BCR::ABL1*-like B-ALL is generally associated with inferior prognosis irrespective of age, MRD-directed therapy may be beneficial in abrogating the prognostic impact.²⁰ In a retrospective multicentric study of the pre-TKI era, *BCR::ABL1*-like ABL-class-r had inferior outcomes.²¹

The genomic risk stratification proposed here was associated with distinct EFS across ICiCLE risk groups and had an independent association with EFS on MVA, in addition to ICiCLE-ALL-14 initial risk and PI-MRD status. On combining genomic data with PI-MRD assessments, we found out that the SGR group was associated with uniformly superior

prognosis irrespective of MRD status. This, however, should not be interpreted as rationale to deescalate therapy as the patients with MRD positive in the SGR category were all treated with high-risk consolidation in this cohort. Therefore, these findings should not be construed as evidence supporting treatment deescalation in MRD-positive SGR patients. MRD-positive status was associated with inferior outcomes within the IGR and HGR categories, suggesting that a combination of genomic risk group and MRD status might form the most optimum strategy for risk-adapted protocols in children with B-ALL. Moorman et al developed the Prognostic Index for UKALL scoring system to stratify patients with pediatric B-ALL based on early treatment response and selected cytogenetic abnormalities. However, their model incorporated a limited set of high-risk genomic alterations, namely *KMT2Ar*, hypodiploidy, *BCR::ABL1*, *TCF3::HLF*, and *iAMP21*.²² Our study evaluates the prognostic impact of a broader spectrum of genomic subtypes as recognized in the recent WHO5 classification. Using a uniformly treated cohort of pediatric B-ALL, we systematically analyzed the clinical relevance of these molecular subtypes to inform a comprehensive, data-driven genomic risk stratification model.

We acknowledge the limitation that patients in our study lack gene expression profiling. Incorporating unbiased sequencing approaches will help further delineate genomic makeup of NOS cases and may identify missed cases. Nevertheless, the stepwise diagnostic workflow, targeted DNA/RNA sequencing strategy, and genomic risk model described in this study represent a potentially practical and scalable approach for routine implementation, particularly in resource-constrained settings. When integrated with MRD assessment, these approaches may contribute to improved risk stratification in childhood B-ALL. However, the clinical applicability of this framework and its incorporation into risk-adapted therapeutic strategies require prospective validation in larger independent cohorts.

In conclusion, we demonstrate applicability of the WHO5 classification in routine practice and create a general framework for incorporation of the WHO5 classification in risk-adapted therapy for childhood B-ALL.

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DATA SHARING STATEMENT

A data sharing statement provided by the authors is available with this article at DOI <https://doi.org/10.1200/GO-25-00665>.

Researchers may request access to anonymized patient data by contacting the corresponding author, Dr Nikhil Patkar, at nvpatkar@gmail.com.

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